

STOR 155.010 Lecture 10

Conditional Probability

Aniruddhan (Ani) Ganesaraman

Fall 2026
UNC Chapel Hill

17 September 2026

1 Computing Probability: Examples

2 Conditional Probability

Computing Probability: Favorable / Total

- For experiments with equally likely outcomes:

$$\mathbb{P}(A) = \frac{\# \text{ of outcomes in } A}{\# \text{ of outcomes in } \Omega}$$

- **Example:** A standard deck of 52 cards is shuffled and one card is drawn at random. What is the probability the card is a face card (Jack, Queen, or King)?

Computing Probability: Favorable / Total

- For experiments with equally likely outcomes:

$$\mathbb{P}(A) = \frac{\# \text{ of outcomes in } A}{\# \text{ of outcomes in } \Omega}$$

- **Example:** A standard deck of 52 cards is shuffled and one card is drawn at random. What is the probability the card is a face card (Jack, Queen, or King)?
 - Total outcomes: $|\Omega| = 52$
 - Favorable outcomes: $3 \text{ face cards} \times 4 \text{ suits} = 12$
 - $\mathbb{P}(\text{face card}) = \frac{12}{52} = \frac{3}{13} \approx 0.231$

Another Example

A jar contains 8 red, 5 blue, and 7 green marbles. One marble is drawn at random. What is the probability it is **not** red?

Another Example

A jar contains 8 red, 5 blue, and 7 green marbles. One marble is drawn at random. What is the probability it is **not** red?

- Total marbles: $8 + 5 + 7 = 20$
- Method 1 (direct count):
favorable outcomes (blue or green) $= 5 + 7 = 12$, so

$$\mathbb{P}(\text{not red}) = \frac{12}{20} = 0.6$$

- Method 2 (complement rule): $\mathbb{P}(\text{red}) = \frac{8}{20} = 0.4$, so

$$\mathbb{P}(\text{not red}) = 1 - 0.4 = 0.6$$

Computing Probability by Simple Counting

Two coins are tossed. What is the probability of getting **at least one** Head?

- List the sample space: $\Omega = \{HH, HT, TH, TT\}$, so $|\Omega| = 4$
- Event $A = \text{"at least one Head"} = \{HH, HT, TH\}$
- $\mathbb{P}(A) = \frac{3}{4}$
- *Alternative:* $\mathbb{P}(A) = 1 - \mathbb{P}(\text{no Heads}) = 1 - \frac{1}{4} = \frac{3}{4}$

Another Example

Two fair six-sided dice are rolled. What is the probability that the sum of the two dice is 8?

- Total outcomes: each die has 6 faces, so $|\Omega| = 6 \times 6 = 36$
- List outcomes summing to 8: $(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)$
- Favorable outcomes: 5
- $\mathbb{P}(\text{sum} = 8) = \frac{5}{36} \approx 0.139$

Practice Example: Survey Data

A university surveyed 1,850 randomly sampled undergraduate students about their preferred study location (Library, Dorm, or Coffee Shop) and whether or not they typically study with music on. 45% of respondents said they prefer the Library, 38% said they typically study with music on, and 20% said both.

- Let L = “prefers the Library” and M = “studies with music on.”
- What is the probability a randomly selected student prefers the Library **or** studies with music on?

Practice Example: Survey Data

A university surveyed 1,850 randomly sampled undergraduate students about their preferred study location (Library, Dorm, or Coffee Shop) and whether or not they typically study with music on. 45% of respondents said they prefer the Library, 38% said they typically study with music on, and 20% said both.

- Let L = “prefers the Library” and M = “studies with music on.”
- What is the probability a randomly selected student prefers the Library **or** studies with music on?

$$\mathbb{P}(L \cup M) = \mathbb{P}(L) + \mathbb{P}(M) - \mathbb{P}(L \cap M) = 0.45 + 0.38 - 0.20 = 0.63$$

- What is the probability a student prefers the Library but does **not** study with music on?

Practice Example: Survey Data

A university surveyed 1,850 randomly sampled undergraduate students about their preferred study location (Library, Dorm, or Coffee Shop) and whether or not they typically study with music on. 45% of respondents said they prefer the Library, 38% said they typically study with music on, and 20% said both.

- Let L = “prefers the Library” and M = “studies with music on.”
- What is the probability a randomly selected student prefers the Library **or** studies with music on?

$$\mathbb{P}(L \cup M) = \mathbb{P}(L) + \mathbb{P}(M) - \mathbb{P}(L \cap M) = 0.45 + 0.38 - 0.20 = 0.63$$

- What is the probability a student prefers the Library but does **not** study with music on?

$$\mathbb{P}(L \setminus M) = \mathbb{P}(L) - \mathbb{P}(L \cap M) = 0.45 - 0.20 = 0.25$$

Law of Large Numbers

- As more observations are collected, the proportion of occurrences with a particular outcome, $\hat{p}_n$, converges to the probability of that outcome, p .
 - If we keep tossing a coin, the proportion of times we observe Tails approximates $1/2$.
 - If we keep throwing a die, the proportion of times we see the number 1 approximates $1/6$.

[Click here for a beautiful simulation!](#)

Overview

1 Computing Probability: Examples

2 Conditional Probability

Motivation: Conditional Probability

- Probability isn't static, new information changes it.
- “What's the chance of rain tomorrow?” vs. “What's the chance of rain tomorrow, *given* it's cloudy today?” have very different answers.
- Medical testing: *overall* disease prevalence $\neq$ chance of disease *given* a positive test.
- **Conditional probability**: updating $\mathbb{P}(A)$ once we know event B occurred.
- The engine behind spam filters, medical diagnosis, weather forecasts, search engines, anywhere new data should change our beliefs.

Example

- Researchers randomly assigned 72 chronic users of cocaine into three groups: desipramine (antidepressant), lithium (standard treatment for cocaine) and placebo. Results of the study are summarized below.

	relapse	no relapse	total
desipramine	10	14	24
lithium	18	6	24
placebo	20	4	24
total	48	24	72

- What is the probability that a patient relapsed?
- What is the probability that a patient took lithium?
- What is the probability that a patient relapsed given that they took placebo?

Conditional Probability

- The **conditional probability** of an event A given an event B is defined to be

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

- This is the probability of A in a world where B has occurred.
- In general, $P(A|B) \neq P(B|A)$.
- Multiplication Rules:

$$P(A \cap B) = P(A|B)P(B) \quad \text{and} \quad P(A \cap B) = P(B|A)P(A)$$

Conditional Probability and Independence

- Events A and B are **independent** if $P(A|B) = P(A)$ and $P(B|A) = P(B)$.
- Events A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

- *Intuition:* the occurrence of B doesn't affect the probability of A , and vice-versa.
- If $P(A|B) \neq P(A)$, A and B are not independent.
- Independence is a symmetric property (A and B are independent if and only if B and A are independent).

Example: Independent Events

- Roll a fair die. $A =$ “roll is even,” $B =$ “roll is 1, 2, 3, or 4.”
- $\mathbb{P}(A) = 3/6 = 0.5$
- $\mathbb{P}(A|B) = \mathbb{P}(\{2, 4\})/\mathbb{P}(\{1, 2, 3, 4\}) = (2/6)/(4/6) = 0.5$
- Since $\mathbb{P}(A|B) = \mathbb{P}(A)$, knowing B occurred tells us **nothing new** about A .
- A and B are **independent**: information about one event doesn't update our belief about the other.

Example: Not Independent

- Draw one card from a standard 52-card deck.
- Let $A =$ “card is a King,” $B =$ “card is a face card (J, Q, K).”
- $\mathbb{P}(A) = 4/52 \approx 0.077$
- $\mathbb{P}(A|B) = \mathbb{P}(A \cap B)/\mathbb{P}(B) = (4/52)/(12/52) = 4/12 \approx 0.333$
- Since $\mathbb{P}(A|B) \neq \mathbb{P}(A)$, knowing B occurred **changes** our belief about A .
- A and B are **not independent**: learning the card is a face card makes it far more likely to be a King.

Properties Conditional Probabilities

- Let $A_1, \dots, A_k$ be all disjoint outcomes. If B is an event,

$$\mathbb{P}(A_1|B) + \dots + \mathbb{P}(A_k|B) = 1$$

- The complement rule also holds when both events are conditioned on the same information:

$$\mathbb{P}(A|B) = 1 - \mathbb{P}(A^c|B)$$

Example

- Classify patient as being on desipramine, lithium, or placebo, and let $B = \text{"patient relapsed."}$ Then

$$\mathbb{P}(\text{Desipramine}|B) + \mathbb{P}(\text{Lithium}|B) + \mathbb{P}(\text{Placebo}|B) = 1$$

- Complement example:** roll a die, let $A = \text{"even,"}$ $B = \text{"roll is at least 3."}$

$$\mathbb{P}(A|B) = \mathbb{P}(\{4, 6\})/\mathbb{P}(\{3, 4, 5, 6\}) = (2/6)/(4/6) = 0.5$$

$$\mathbb{P}(A^c|B) = \mathbb{P}(\{3, 5\})/\mathbb{P}(\{3, 4, 5, 6\}) = (2/6)/(4/6) = 0.5$$

Motivating Bayes' Theorem

- Conditional probability lets us compute $\mathbb{P}(B|A)$.
Bayes' Theorem flips the direction: it connects $\mathbb{P}(B|A)$ into $\mathbb{P}(A|B)$.
- **Example.** Your roommate texts “I definitely failed the exam.”
Historically, people who say this pass 80% of the time. People who stay quiet pass 95% of the time. How worried should they actually be?
- Bayes' Theorem: the tool for turning gut-reaction dread into an actual number.